When Judgement Does Not Stay the Same

Stability and Variation in Repeated AI Evaluation

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Executive Summary

AI systems are often evaluated through single outputs, with reliability inferred from whether a response appears reasonable, unbiased, or well explained. However, this approach provides limited insight into how judgements behave when the same task is repeated.

This paper examines judgement stability in repeated evaluations of recruitment text. While most classifications remain consistent across runs, a meaningful proportion vary under identical conditions. These variations are not random, but concentrated at decision boundaries particularly between adjacent categories where interpretation is required.

The findings suggest that instability in AI judgement is not best understood as error, but as a structured property of interpretation under uncertainty. Evaluation approaches that rely on single outputs may therefore overlook important aspects of system behaviour.

Abstract

Most approaches to evaluating AI systems focus on the quality of individual outputs. However, single observations provide limited insight into how systems behave when the same task is repeated. Building on the Behavioural Evaluation Framework (Hull, 2026), this paper examines judgement stability under repeated execution.

Using a series of repeated evaluations of job advertisements for potential age bias, the analysis explores how classification outcomes vary under identical conditions. The findings indicate that variation is not random but concentrated at decision boundaries, particularly between adjacent categories such as “Potentially Biased” and “Unclear”.

These observations suggest that instability in AI judgement reflects structured sensitivity to interpretation rather than isolated error. The paper argues that reliability in AI judgement systems is better understood through patterns of behaviour across repeated evaluations than through the inspection of individual outputs.

Keywords

- AI evaluation
- LLM evaluation
- model reliability
- judgement stability
- behavioural audit
- decision boundaries

1. Introduction

Evaluation of AI systems often rests on the inspection of individual outputs. A response that appears reasonable is frequently taken as evidence that the system is behaving reliably.

This assumption becomes less certain when the same task is observed repeatedly.

Language-based AI systems generate responses probabilistically. Where interpretation is required, the judgement produced in one instance does not necessarily remain the same when the task is executed again. A single output may appear coherent and complete, but it does not reveal whether the underlying judgement is stable. This paper examines where that stability holds, and where it begins to give way.

In many fields that deal with complex systems, reliability is assessed through repeated operation rather than isolated observation (Ebeling, 2019; Leveson, 2016).

Building on the Behavioural Evaluation Framework introduced in earlier work (Hull, 2026), which proposes evaluating AI systems through repeated execution, internal signals, and independent system comparison, this paper focuses on one specific aspect of behaviour: the stability of judgement under repeated execution.

2. Research Context

This paper forms part of the *Agents at Work* research series, which examines how AI systems interpret age-coded language in recruitment text. Earlier phases explored detection and reasoning patterns across a large dataset of job advertisements, while subsequent work introduced repeated evaluation as a method for observing behavioural properties.

The Behavioural Evaluation Framework positioned repeated execution as a means of revealing patterns that remain invisible in single outputs. Initial results suggested that while many judgements remain stable, others vary under identical conditions.

These findings draw on the Phase 4 behavioural evaluation study conducted within the *Agents at Work* research series, which examined judgement stability across repeated evaluations of a structured sample of recruitment advertisements.

This paper examines that variation directly, focusing on its structure and distribution.

3. Method Overview

The analysis is based on repeated evaluations of job advertisements for potential age bias. Each advertisement was assessed multiple times using the same prompt and model configuration.

The task required classification into one of four categories:

- Likely Biased
- Potentially Biased
- Unclear
- Unbiased

By holding input and prompt conditions constant, the analysis isolates variation arising from the model’s internal processes rather than external changes.

The focus of the analysis is not classification accuracy, but the stability of judgement across repeated executions.

4. Findings

4.1 Overall Stability

Across the sample, the majority of evaluations produced consistent outcomes. Approximately four-fifths of advertisements received the same classification across all runs.

However, a meaningful minority did not.

In roughly one-fifth of cases, classifications varied when the same advertisement was evaluated repeatedly under identical conditions.

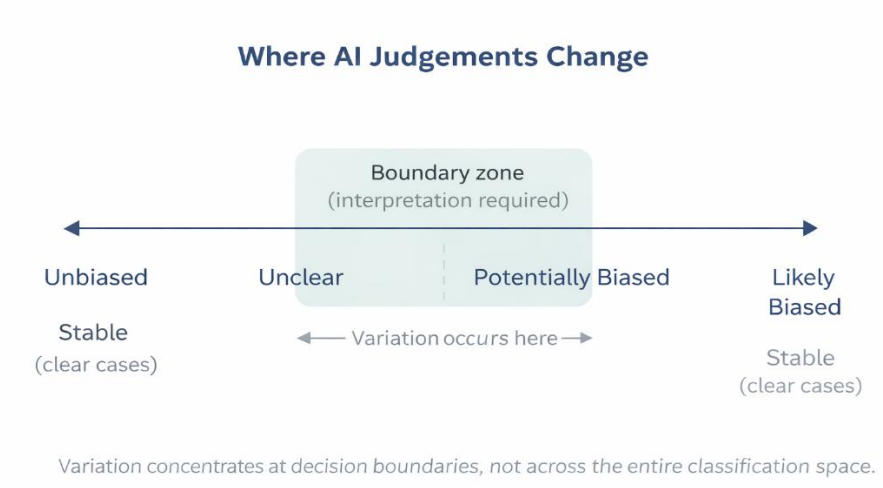
4.2 Distribution of Variation

Variation was not evenly distributed across categories.

Most changes occurred between the categories “Potentially Biased” and “Unclear”. Movement into or out of the more definitive categories “Likely Biased” and “Unbiased” was comparatively rare.

This indicates that variation is concentrated within a narrow region of the classification space rather than occurring randomly across all outcomes.

Figure 1 – Where AI judgements change under repeated evaluation



Variation tends to appear in a narrower region where interpretation is required, rather than across the full range of outcomes.

4.3 Example of Boundary-Level Variation

A single advertisement describing an early-career role illustrates this pattern.

In one evaluation, the model classified the text as “Unclear”, noting that while certain phrases could imply early-career targeting, the evidence was not explicit. In a subsequent run using

the same input, the classification changed to “Potentially Biased”, with a similar rationale but a slightly stronger interpretation of the same cues.

The explanation remained broadly consistent across both runs. The system identified similar features and articulated comparable reasoning. The change occurred at the point where interpretation was translated into a categorical decision.

5. Interpretation

The observed variation does not suggest that the system is behaving erratically. Instead, it indicates that the system is operating at a decision boundary.

Where signals are clear and unambiguous, judgements tend to remain stable across repeated evaluations. Where signals are weaker or open to interpretation, the system may move between adjacent categories without any change in input.

This behaviour reflects sensitivity to marginal differences in internal reasoning processes rather than a failure to recognise underlying cues.

Importantly, the explanations accompanying these judgements often remain plausible even when classifications change. This creates the appearance of consistency at the level of individual outputs while variation persists across repeated observation.

6. Implications for AI Evaluation

These findings have implications for how AI systems are assessed in practice.

Evaluation methods that rely on single outputs risk overlooking structured instability at decision boundaries. A system may appear reliable when observed once, while exhibiting variation when examined across repeated executions.

In contexts where AI systems contribute to decision-making, such variation may influence how borderline cases are interpreted. Small differences in judgement at the boundary may lead to different outcomes when applied at scale.

These observations suggest that reliability in AI judgement systems cannot be assessed solely through isolated outputs. It requires examination of how judgements behave across repeated evaluations.

7. Limitations

This study focuses on a specific evaluative task involving recruitment text and age-coded language. While the observed patterns are consistent with broader behavioural expectations of probabilistic systems, further work is required to examine whether similar patterns appear across other domains.

The analysis is based on repeated evaluation under controlled conditions and does not examine how behaviour may evolve over time or under changing prompts.

Future work will extend this approach through controlled perturbation and stress-testing techniques to examine how judgement stability responds to variations in input and context.

8. Conclusion

When the same evaluative task is repeated, AI judgements do not always remain the same. This variation is not random, but structured. It emerges most clearly at decision boundaries where interpretation is required.

Understanding this behaviour requires a shift in perspective. Rather than assessing whether a single output appears correct, evaluation must consider whether the underlying judgement holds across repeated observation.

Repetition does not introduce instability. It reveals whether a judgement is stable.

References

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Citation

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